9. Common Mistakes in Quant Strategy Development

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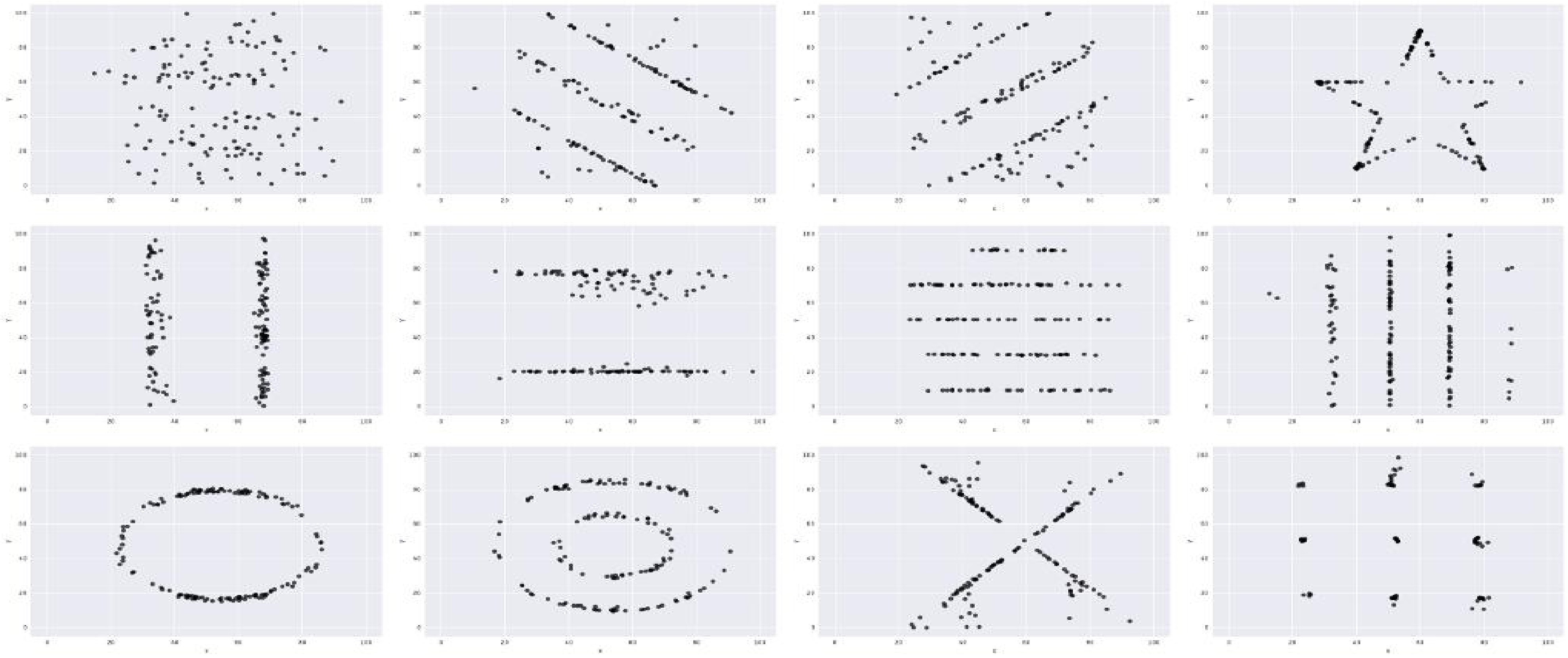
1. Common Mistakes: Data

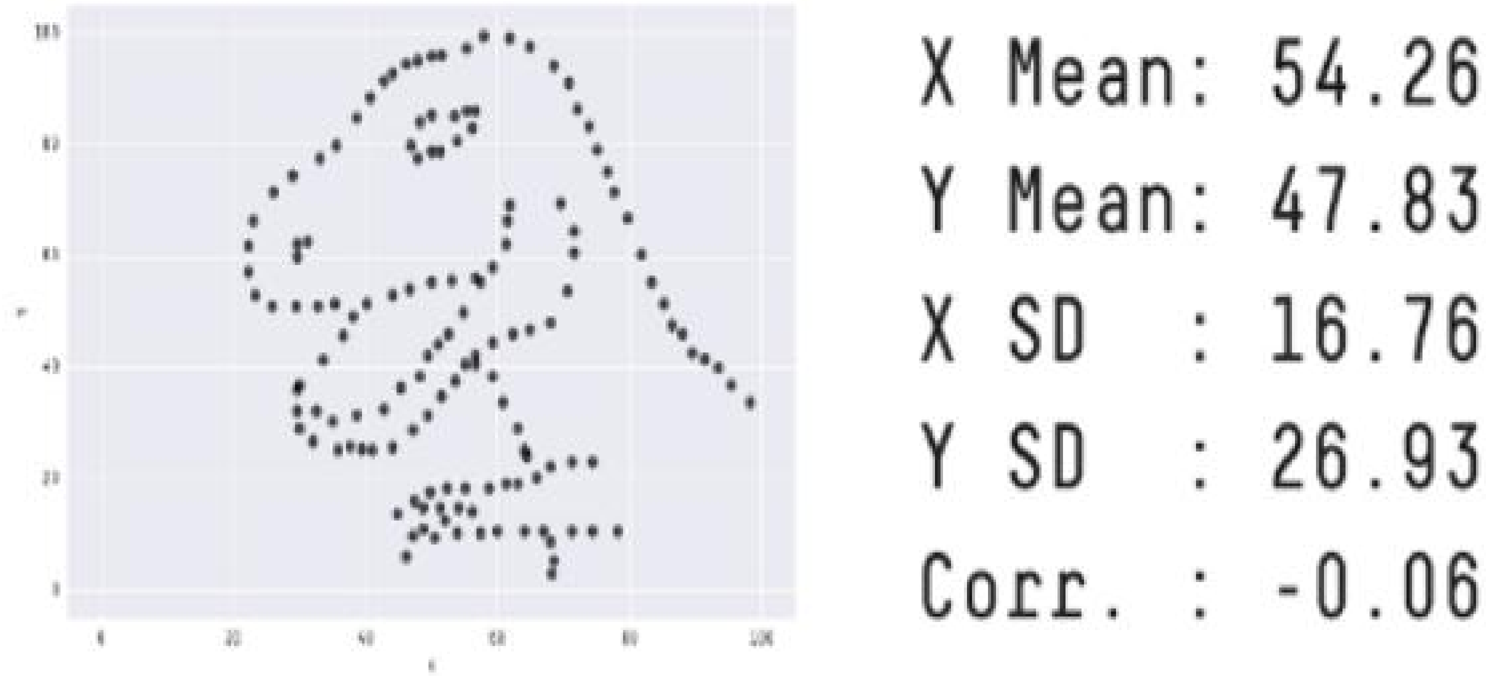
- **Mistake 1**: μ and σ are not enough.

Simply computes μ and σ, inputs them into the model, and obtains results without examining the dataset’s underlying features.



Matejka, J., & Fitzmaurice, G. (2017, May). Same stats, different graphs: generating datasets with varied appearance and identical statistics through simulated annealing. In *Proceedings of the 2017 CHI conference on human factors in computing systems* (pp. 1290-1294).

1. Common Mistakes: Data



- The T-rex has the same μ and σ with all the data showed in the previous page!
- In a 1-D scenario, the patterns are much more difficult to be recognized.
- In Finance, the parameters μ (expected return) and σ (standard deviation) quantify the fundamental trade-off between risk and reward. They are the inputs of mean-variance analysis, Black-Schole’s formula, factor models, etc.

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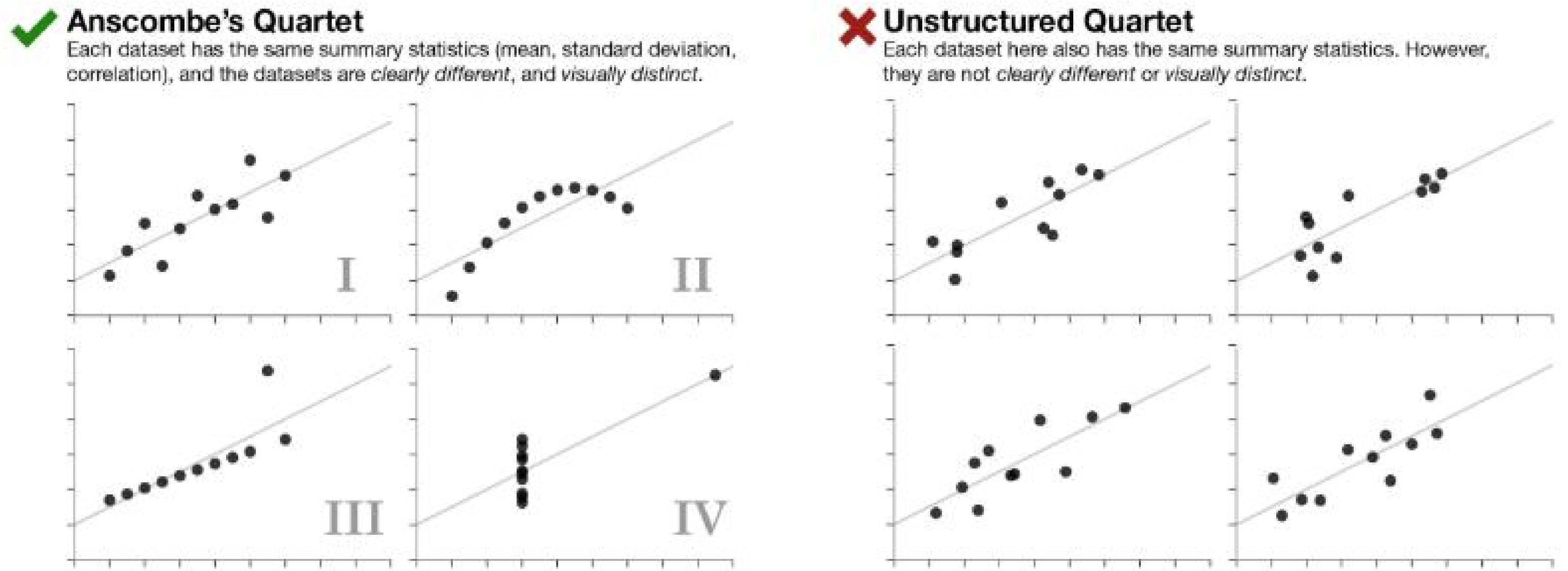
1. Common Mistakes: Data

- **Mistake 2**: Statistically significant is not enough.

Run the regression and check the p-values for significance, assuming datasets are the same with the same β.



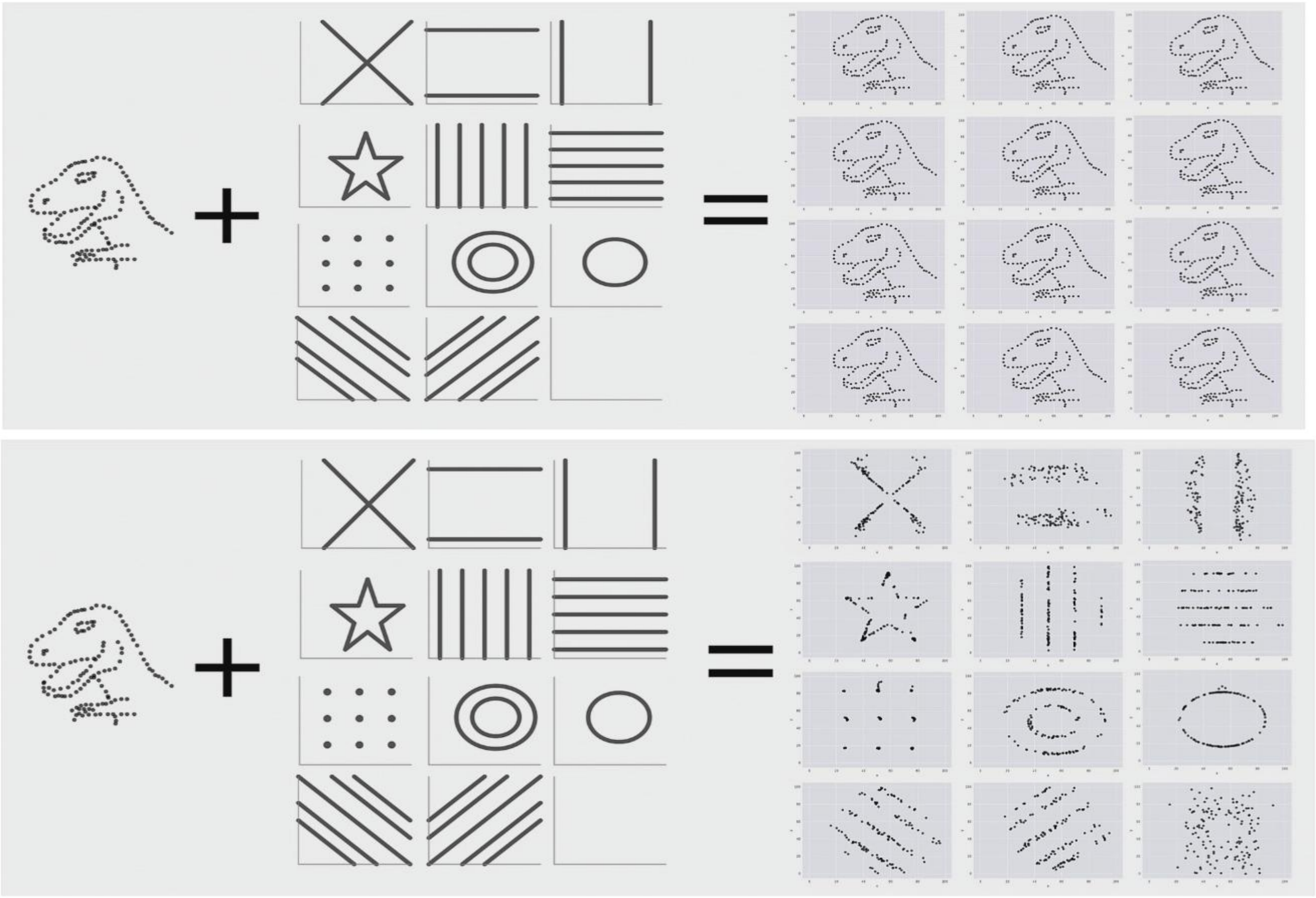
https://www.research.autodesk.com/publications/same-stats-different-graphs/

1. Common Mistakes: Data

- **Mistake 3 (Data)**: Correlation is not enough.

Your raw data is not just a signal+ a noise. It could be a T-rex+a star, while statistically, μ and σ stay the same and even the correlation is ZERO between the T-rex and star.

- **Conjecture**: Factor models are vulnerable to this problem.



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2. Common Mistakes: Modeling

- **Mistake 4**: **Missing the underlying logic**, only rely on statistical results.

Conjecture： AI models are missing underlying logic for developing a trading strategy.

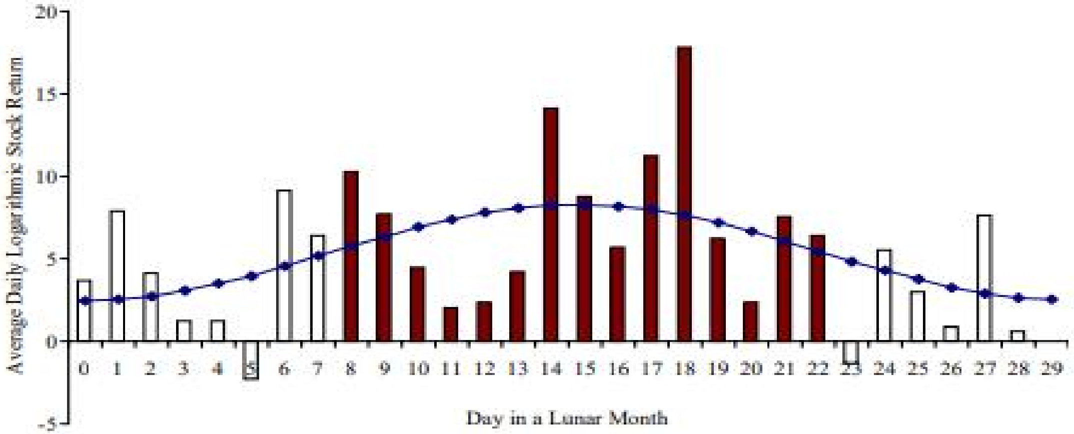


Fig. 2. Average daily logarithmic return of the global portfolio by lunar dates. This figure graphs, for each day of a lunar month, the average daily logarithmic stock returns of an equal-weighted global portfolio of the 48 country stock indices. Day 0 is a full moon day and day 15 is around a new moon day (day 15 is around new moon day since the length of a lunar month varies). The curved line is the estimated sinusoidal model of the lunar effect on stock returns from the following estimated equation: $R_t = 5.38 - 2.88 * \cos(2\pi d/29.53)$, where d is the number of days since the last full moon.

Lunar phases and stock returns: joint tests

	Lunar dummy (β)	
	15-Day window	7-Day window
G7	−2.43 (−1.53)	−2.45 (−1.09)
Other developed markets	−3.21** (−2.08)	−3.19 (−1.47)
Emerging markets	−5.60** (−2.11)	−11.27*** (−3.12)
All markets	−3.95** (−2.25)	−5.93** (−2.44)
All markets with country group dummies	−3.94** (−2.24)	−5.92** (−2.43)
All markets with country fixed effects	−4.63*** (−3.54)	−8.19*** (−3.59)

This table reports the estimates of a pooled regression with panel corrected standard errors (PCSE): $R_{it} = \alpha_i + \beta * \text{Lunardummy}_t + e_{it}$ for the 15-day window and 7-day window, respectively, where R_{it} is average daily logarithmic returns for country i in lunar month t for each full moon and new moon period. Lunardummy is a dummy variable equal to one if it is a full moon period and zero if it is a new moon period. We define the full moon and the new moon periods using the 15-day and the 7-day windows. In the 15-day window specification, we define the full moon period as 7 days before and after the full moon day plus the full moon day, and define the new moon period as the rest of the lunar month. In the 7-day window specification, we define the full (new) moon period as 3 days before and after the full (new) moon day plus the full (new) moon day. The PCSE specification adjusts for the contemporaneous correlation and heteroscedasticity among country indices and for the autocorrelation within each country's stock index. T-statistics are reported in the parentheses. The daily returns are in basis points.

*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively, using a two-tailed test.

Yuan, K., Zheng, L., & Zhu, Q. (2006). Are investors moonstruck? Lunar phases and stock returns. *Journal of Empirical Finance*, *13*(1), 1-23.

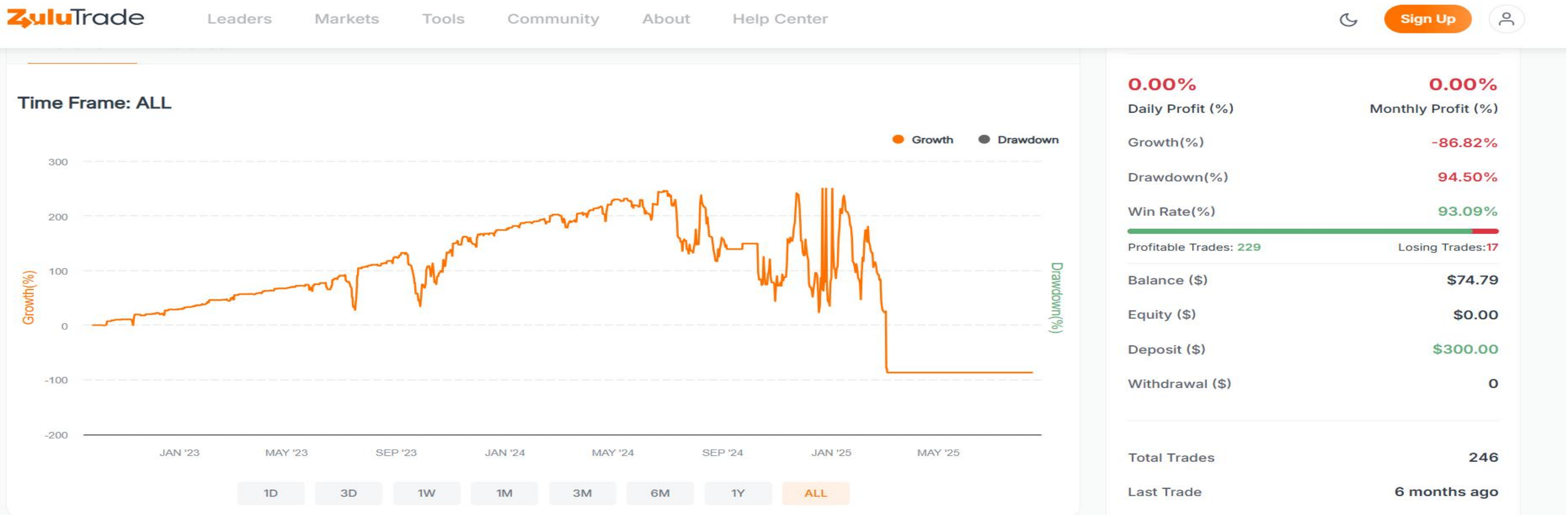
2. Common Mistakes: Modeling

- **Mistake 5:** Too Tight Stoploss/Takeprofit

Setting too tight stop-loss (SL) or take-profit (TP) levels is a common but critical error in trading strategy design. It often stems from an overly cautious or unrealistic view of market behavior.

A tight take-profit will significantly increase the winning probability. However, even with a win probability above 90%, the strategy can still lose money if a single losing trade outweighs the gains from dozens of winning trades.

A tight stop-loss tends to be triggered frequently, which can lead to substantial drawdowns driven by the accumulation of many small losses.



2. Common Mistakes: Modeling

- **Mistake 6**: Overfitting

Overfitting refers to the process of creating a trading model that is excessively tailored to past market data (historical noise and random patterns) rather than capturing genuine, persistent market inefficiencies.

Overfitting Mistakes	Example	Live Trading Outcome
Excessive Parameters	A strategy with 10 tunable parameters and an "optimal" combination found through grid search	The strategy fails as soon as parameters deviate slightly
Look-Ahead Bias	Unintentional use of the close price for modeling, assumming buy at open.	The strategy appears profitable in backtests but is impossible to execute live
Coincidental Patterns	"Discovering" that gold always falls the day after the S&P 500 rises on a Tuesday during a full moon	The pattern lacks logic and will not recur in the future
Overfitting through Factor Mining	Testing 10,000 factors and selecting the 1 with the best historical performance	The selected factors exhibit only spurious correlation

2. Common Mistakes: Modeling

- **Mistake 7:** More is not Better

Combining Multiple Signals Doesn’t Necessarily Improve the Performance:

Signal Fusion	Example	Live Trading Outcome
Correlated Signals	With specific parameters, using RSI and MACD together may add little value, as their signals are highly correlated.	No improvements to the performance
Same Logic, Different Parameters	Combining RSI(5) and RSI(30) may underperfrom than RSI(5) itself by filtering out numbers of profitable trades.	It may or may not work. Need backtest to validate.
Too Many Signals	Combining RSI(5), MA(30), MA(60), ADX(30), ..., may filter out most of trading opportunities.	Optimization algorithms, machine learnings may help. However, it remains essential to be clear about the purpose of combining them.
Overfitting	Input 1000 technical indicators to a deep learning algorithm and get the final decision.	Combining n signals with optimization algorithms is actually introducing n additional parameters to your strategy. Excessive parameterization, especially when optimized, can easily lead to overfitting.

2. Common Mistakes: Modeling

- **Mistake 8**: Lack of Drawdown Management

Lack of Drawdown Management refers to the failure to actively control, monitor, and plan for portfolio losses (drawdowns) within a quantitative trading strategy.

Loss	Required Return for Recovery	Difficulty Analogy	Theoretical Recovery Time (at 10% annual return, uninterrupted)
10%	11.1%	Can be recovered with slight outperformance.	~1 year
20%	25%	Requires significant outperformance vs. benchmark (e.g., long-term global equity average ~9% annualized).	~2.5 years
30%	42.9%	Comparable to the annualized return of top-tier hedge funds (e.g., Soros, Renaissance).	4-5 years
50%	100%	Requires a doubling of capital (extremely difficult in the short term).	~7.2 years
70%	233%	Nearly irreversible capital loss.	~9+ years

2. Common Mistakes: Modeling

Even with quant strategies, senarios of loss are also inevitable. We need a plan to deal with loss such as diversification, hard stop-loss, dynamic performance monitoring with kick-out rules, etc.

Case Study	Index / Asset	Max Drawdown	Drawdown Period	Required Gain for Recovery	Actual Recovery Time	Recovery Period
2000 Dot-com Bubble	Nasdaq 100	-83%	Mar 2000 - Oct 2002	+488%	15 yrs	Oct 2002 - Apr 2017
2008 Global Financial Crisis	S&P 500	-57%	Oct 2007 - Mar 2009	+133%	4 yrs, 3 mo	Mar 2009 - Jun 2013
2015 Chinese Market Circuit Breaker	SSE Index	-70%	Jun 2015 - Feb 2016	+233%	5 yrs	Feb 2016 - Feb 2020
2018 Crypto Winter	Bitcoin (BTC)	-84%	Dec 2017 - Dec 2018	+525%	3 yrs, 2 mo	Dec 2018 - Feb 2021

2. Common Mistakes: Programming

- **Mistake 9**: Misuse of functions.

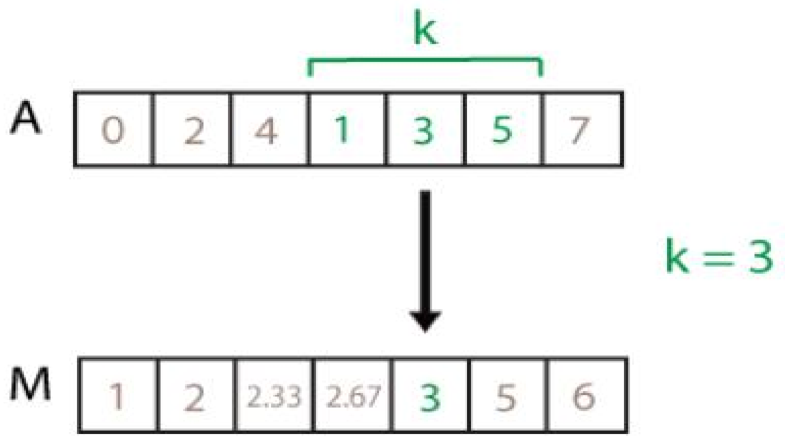
Use a function before carefully examine the code. Sometime you may use the data from the future for modeling the current!

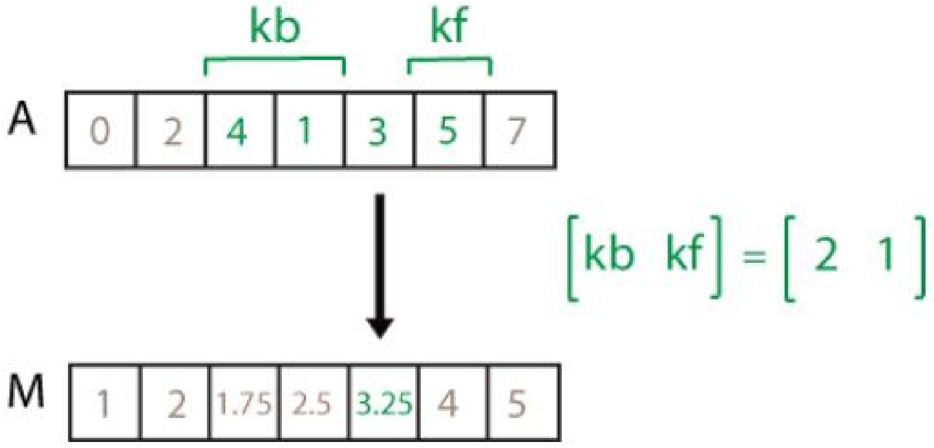
My own mistake:
In Matlab, a function can calculate moving average is
M=movmean(A, k)
where k is the rolling window.

Since it's not specifictly defined for time series, the function default is to calculate the centered average includes the element in the current position plus surrounding neighbors.

When processing with time series, the centered average is using k/2-1 data in the **FUTURE!** It will make the backtest looks too good to be true!

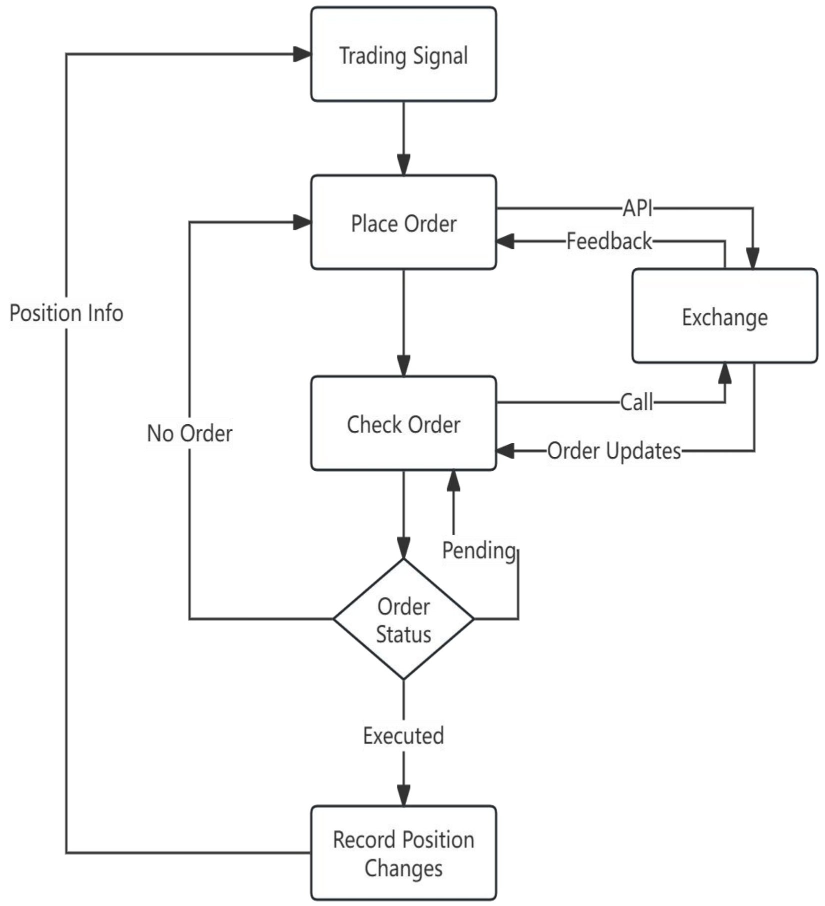
In fact, the trailing moving average of a time series should be written as
M = movmean(A,[k 0])
meaning using 0 elements forward.





2. Common Mistakes: Programming

- **Mistake 10**: Flaws of the System



Scenario 1: If an exchange's confirmation of a newly placed order is delayed or lost, a subsequent "check order status" request may incorrectly report that the order does not exist. An automated system interpreting this as a failure could then place a duplicate order, resulting in unintended double the intended position.

Scenario2: If an exchange's market data feed is delayed by 10 seconds, any timer-based logic in your trading system will be processing severely stale data. This will cause calculations, such as indicator values, volatility estimates, or arbitrage signals, to be generated from incorrect, outdated price and order book information, leading to erroneous trading decisions.

Scenario 3: If a trading system generates a new buy signal before it has finished processing and updating its internal state from the previous signal, a race condition occurs. It is a critical state synchronization flaw in trading system architecture. The system’s signal generation module and position management module fall out of sync, causing the system to "lose track" of its true market exposure.

Thank You

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